assignemnt_three

2024-04-25

#libraries

```
library(tidyverse)
library(caret)
library(moments)
library(inspectdf)
library(rsample)
library(modelr)
library(car)
library(tidymodels)
```

```
#student number
ysn = 1895211
#determining the file number
filenum <- (ysn + 2) %% 3
filenum
```

```
## [1] 2
```

```
filename <-paste0("./data/survey_",filenum,".csv")
filename
```

```
## [1] "./data/survey_2.csv"
```

#Q1 Load the correct dataset and save it as a tibble. Output the first 10 lines of the dataset. #and #Q3 Now it's time to tame our data. But since we are going to fit a logistic regression model, we need to modify our requirements a little bit. • Make sure that all column names are in snake case. • Makethe variables age, company aware, malfunction, multi purch and social media conform to the Tame Data conventions in Module 2 (page 3). • Convert recommend to a data type, with yes for 1 and no for 0. • Convert the Socio-Economic Status to a

```
#Reading data set and taming the data
dataset = read_csv(filename)

colnames(dataset) <- str_to_lower(colnames(dataset))
dataset$age <- as.integer(dataset$age)

#making company aware, malfunction, multi purch and social media as logical
dataset$company_aware <- as.logical(dataset$company_aware)
dataset$malfunction <- as.logical(dataset$malfunction)
dataset$multi_purch <- as.logical(dataset$multi_purch)
dataset$social_media <- as.logical(dataset$social_media)
```

```r
# Convert recommend to a <fct> data type, with yes for 1 and no for 0.
dataset$recommend <- as.factor(dataset$recommend)
dataset$recommend <- fct_recode(dataset$recommend, "yes" = "1" , "no" = "0")

#Convert the Socio-Economic Status to a <fct>
dataset$ses <- as.factor(dataset$ses)

head(dataset,10)
```

```
## # A tibble: 10 x 7
##    recommend   age company_aware malfunction multi_purch ses   social_media
##    <fct>     <int> <lgl>         <lgl>       <lgl>       <fct> <lgl>
##  1 no           49 TRUE          FALSE       FALSE       mid   TRUE
##  2 no           46 TRUE          TRUE        TRUE        low   FALSE
##  3 yes          50 TRUE          FALSE       TRUE        mid   FALSE
##  4 no           37 FALSE         TRUE        FALSE       mid   TRUE
##  5 no           31 FALSE         FALSE       FALSE       high  TRUE
##  6 yes          45 FALSE         FALSE       FALSE       low   FALSE
##  7 no           26 TRUE          TRUE        TRUE        mid   TRUE
##  8 no           29 TRUE          FALSE       FALSE       low   TRUE
##  9 no           64 FALSE         FALSE       FALSE       high  FALSE
## 10 yes          47 TRUE          FALSE       FALSE       mid   TRUE
```

Q2 Using dot points, identify what types of variables we now have in our data set, i.e., "Quantitative Discrete", "Quantitative Continuous", "Categorical Nominal", "Categorical Ordinal". (Don't just describe what data type they are in the tibble — you need to think about the type of variable in the context of the meaning of the data.) Make sure you provide some justification for your choice of variable types.

-recommend (Categorical Nominal): This variable is categorical because it takes on two distinct categories (0 and 1), which tells us Gadget was recommended or not. It's nominal as there is no specific order between these two categories.

-age (Quantitative Discrete): Age is measured in whole years, making it a discrete quantitative variable because it involves counting of whole (years).

-company aware (Categorical Nominal): This is a binary variable (TRUE/FALSE), categorizing people by whether they know the company name. no specific order, making it nominal.

-multi purch (Categorical Nominal): This variable indicates whether more than one Gadget was purchased (TRUE/FALSE). Like 'company aware', it is nominal as it divides the respondents into groups without any order.

-malfunction (Categorical Nominal): This indicates whether any of the Gadgets malfunctioned (TRUE/FALSE). It's nominal since it categorizes gadgets into malfunctioning or not.

-SES (Categorical Ordinal): Socio-Economic Status is recorded as high, mid, or low. This is a categorical variable with a clear ranking order, hence it is ordinal.

-social media (Categorical Nominal): This variable categorizes people based on whether they are active on social media (TRUE/FALSE). It's nominal because no specific order.

#4. Setting the correct seed, split your data into a training set (with 40,000 rows) and a testing set, with the remaining rows. Use the command dim() to output the dimensions of your training and testing sets.

```
set.seed(ysn)

data_split <- initial_split(dataset,prop=0.8)
data_train = training(data_split)
data_test = testing(data_split)

dim(data_train)
```

```
## [1] 40000     7
```

```
dim(data_test)
```

```
## [1] 10000     7
```

5. Fit a logistic regression model to your training data, with recommend as the response and all other variables as the predictors. Output the summary of the model.

```
lgr <- logistic_reg() %>%
  set_engine("glm")
lrfit <- lgr %>%
  fit(recommend ~., data= data_train)

summary(lrfit$fit)
```

```
##
## Call:
## stats::glm(formula = recommend ~ ., family = stats::binomial,
##     data = data)
##
## Coefficients:
##                     Estimate Std. Error z value Pr(>|z|)
## (Intercept)         0.094851   0.075322   1.259   0.2079
## age                -0.051856   0.001390 -37.300   <2e-16 ***
## company_awareTRUE  -0.030847   0.030043  -1.027   0.3045
## malfunctionTRUE    -5.753987   0.189183 -30.415   <2e-16 ***
## multi_purchTRUE     3.180904   0.032862  96.796   <2e-16 ***
## seslow              0.329906   0.035785   9.219   <2e-16 ***
## sesmid             -0.006796   0.036611  -0.186   0.8527
## social_mediaTRUE   -0.072315   0.039998  -1.808   0.0706 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 48075  on 39999  degrees of freedom
## Residual deviance: 29663  on 39992  degrees of freedom
## AIC: 29679
##
## Number of Fisher Scoring iterations: 8
```

6. Use the command model matrix() on the ses variable of your training data to see what happens to ses when we fit a model. (See pages 2– 6 in Module 7.)

```
mm <- model_matrix(data_train,~ses)
mm
```

```
## # A tibble: 40,000 x 3
##    `(Intercept)` seslow sesmid
##            <dbl>  <dbl>  <dbl>
##  1             1      0      0
##  2             1      0      0
##  3             1      0      1
##  4             1      0      1
##  5             1      0      0
##  6             1      1      0
##  7             1      0      0
##  8             1      1      0
##  9             1      0      1
## 10             1      1      0
## # i 39,990 more rows
```

#(a) How many new variables have been introduced? - 2 new variables have been introduced after running the command model_matrix(): seslow & sesmid

#(b) What is the reference level for ses? - as the missing variable here is seshigh,seslow,sesmid if seshigh and it is the reference level.

#Q7 (a) Build a new tibble called ses matrix, with the first column giving the true ses data, and the second and third columns giving the coordinates of the ses value in the new variables defined for the ses variable. Call these new variables seslow and sesmid.

```
ses_matrix = tibble( ses = data_train$ses, seslow = mm$seslow, sesmid = mm$sesmid)
ses_matrix
```

```
## # A tibble: 40,000 x 3
##    ses   seslow sesmid
##    <fct>  <dbl>  <dbl>
##  1 high       0      0
##  2 high       0      0
##  3 mid        0      1
##  4 mid        0      1
```

```
##  5 high       0      0
##  6 low        1      0
##  7 high       0      0
##  8 low        1      0
##  9 mid        0      1
## 10 low        1      0
## # i 39,990 more rows
```

#(b) With the coordinates of the form (seslow, sesmid) use the ses matrix and/or the information from Question 6 to write down the coordinates of the ses levels "high", "mid" and "low" in terms of these new variables.

high : 0,0 mid : 0,1 low : 1,0

#Q8 Since we are using general linear models, the model summary describes linear geometric objects, where the dimension of the geometric object is determined by the number of continuous predictors. We have only a single continuous predictor so our model describes a set of lines. How many lines are described by the modelin Question 5? Make sure you give some justification for your answer.

There are 48 lines described by the model : company_aware(t/f), malfunction(t/f), multi_purch(t/f), social_media(t/f), ses(high,mid,low)

so 2 x 2 x 2 x 2 x 3 = 48, as each predictor has 2 values with true false

#Q9 Now it is time to get serious with our data. There may be some interactions between the variables in the data set, so fit a new model to your training set using all the individual variables and all the second-order interaction terms. Use Anova() to find the p-values for each of the variables. Identify all interaction terms that meet the 99.9% significance level.

```
lgr_two <- logistic_reg() %>%
  set_engine("glm")
lrfit_two <- lgr_two %>%
  fit(recommend ~.^2, data= data_train)

#Anova()
Anova(lrfit_two$fit)
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: recommend
##                             LR Chisq Df Pr(>Chisq)
## age                           1542.5  1  < 2.2e-16 ***
## company_aware                    0.9  1    0.34144
## malfunction                   5791.7  1  < 2.2e-16 ***
## multi_purch                  12907.6  1  < 2.2e-16 ***
## ses                            115.2  2  < 2.2e-16 ***
## social_media                     2.9  1    0.08774 .
## age:company_aware                2.4  1    0.12060
## age:malfunction                 21.3  1  3.861e-06 ***
## age:multi_purch                  0.1  1    0.73869
## age:ses                          0.5  2    0.78260
## age:social_media                 1.1  1    0.29573
## company_aware:malfunction        4.4  1    0.03517 *
## company_aware:multi_purch        0.7  1    0.41589
## company_aware:ses                0.5  2    0.78657
## company_aware:social_media       0.3  1    0.60060
```

```
## malfunction:multi_purch         0.1  1    0.76411
## malfunction:ses                 0.0  2    0.97588
## malfunction:social_media        1.2  1    0.26802
## multi_purch:ses                41.3  2  1.064e-09 ***
## multi_purch:social_media        1.0  1    0.32293
## ses:social_media                1.0  2    0.61887
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

#Identifying 99.9% age:malfunction multi_purch:ses

#10 #10a. So first fit a new model with just the individual variables and the significant interactions terms that you identified in Question 9. Show the Anova() output.

```
lgr_three <- logistic_reg() %>% set_engine("glm")
lrfit_three <- lgr_three %>%
  fit(recommend ~.+age:malfunction+multi_purch:ses, data = data_train)

Anova(lrfit_three$fit)
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: recommend
##                 LR Chisq Df Pr(>Chisq)
## age               1541.4  1  < 2.2e-16 ***
## company_aware        0.9  1    0.33814
## malfunction       5845.4  1  < 2.2e-16 ***
## multi_purch      12912.8  1  < 2.2e-16 ***
## ses                115.7  2  < 2.2e-16 ***
## social_media         3.1  1    0.08042 .
## age:malfunction     19.5  1  1.016e-05 ***
## multi_purch:ses     46.9  2  6.502e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

#10b.Then continue with the proper step-by-step backwards stepwise regression to find a model where all terms (individual terms and interaction terms) meet the 95% significance level. At each step, identify the variable that you will remove, and why you will choose that one. Then show the resulting Anova() after you fit each model.

```
bsr <- logistic_reg() %>% set_engine("glm")
bsfit <- bsr %>%
  fit(recommend ~.+age:malfunction+multi_purch:ses, data = data_train)

Anova(bsfit$fit)
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: recommend
##                 LR Chisq Df Pr(>Chisq)
## age               1541.4  1  < 2.2e-16 ***
## company_aware        0.9  1    0.33814
## malfunction       5845.4  1  < 2.2e-16 ***
```

```
## multi_purch      12912.8  1  < 2.2e-16 ***
## ses                115.7  2  < 2.2e-16 ***
## social_media         3.1  1    0.08042 .
## age:malfunction     19.5  1  1.016e-05 ***
## multi_purch:ses     46.9  2  6.502e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

#We will remove company_aware as it is the largest p_value and also >0.05.

```
#refit
bsr_1 <- logistic_reg() %>% set_engine("glm")
bsfit_1 <- bsr_1 %>%
  fit(recommend ~age+malfunction+multi_purch+ses+social_media+age:malfunction+multi_purch:ses, data = da

Anova(bsfit_1$fit)
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: recommend
##                 LR Chisq Df Pr(>Chisq)
## age               1541.2  1  < 2.2e-16 ***
## malfunction       5844.7  1  < 2.2e-16 ***
## multi_purch      12912.1  1  < 2.2e-16 ***
## ses                115.6  2  < 2.2e-16 ***
## social_media         3.1  1    0.07989 .
## age:malfunction     19.5  1  1.022e-05 ***
## multi_purch:ses     47.1  2  6.032e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

We will remove social_media as it is greater than 0.05 and Final Model

```
#FINAL MODEL
bsr_2 <- logistic_reg() %>% set_engine("glm")
bsfit_2 <- bsr_2 %>%
  fit(recommend ~age+malfunction+multi_purch+ses+age:malfunction+multi_purch:ses, data = data_train)

Anova(bsfit_2$fit)
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: recommend
##                 LR Chisq Df Pr(>Chisq)
## age               1916.3  1  < 2.2e-16 ***
## malfunction       5844.4  1  < 2.2e-16 ***
## multi_purch      12909.4  1  < 2.2e-16 ***
## ses                115.6  2  < 2.2e-16 ***
## age:malfunction     19.6  1  9.796e-06 ***
```

```
## multi_purch:ses     47.2  2  5.655e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

#Q11(a) Which interaction terms are significant in your final model? age:malfunction and multi_purch:ses are the significant interaction terms

b.Thinking about the context of the data, provide some reasonable hypotheses for why those interaction terms might represent real effects (and are not just statistical noise).

age:malfunction - Older people might want the gadget to be rough and tough and expectations regarding malfunctions of it.Those expectations might result in negative feedback while younger people are more tech-savy they might keep in mind that,as a new product it might malfunction and also might depend on how they handle it.

multi_purch:ses - People make multipe purchases might be more with the people who have a higher SES, as have would be having more financial freedom on that aspect.And there might be people who prefer this brand over other and are loyal customers who buy it multiple times, so the ses with multiple puchases might be stronger.

#Q12

$$\hat{f_i} = \beta_0 + \beta_1 x_i + ... + \beta_{ip} x_{ip} + \epsilon_i$$

$$\hat{f_i} = \hat{\beta}_0 + \hat{\beta}_1 age + \hat{\beta}_2 malfunction + \hat{\beta}_3 multi_purch + \hat{\beta}_4 seslow + \hat{\beta}_5 sesmid + \hat{\beta}_6 age : malfunction + \hat{\beta}_7 multi_purch : seslow + \hat{\beta}_8$$

#Looking at Question 12, the geometric situation is slightly more complicated now than in Question 8, although our model should still produce a set of lines.

#(a) How many lines does your final model describe? Make sure you provide some justification for your answer.

Without interactions, the combinations of malfunction, multi_purch, seslow, sesmid would produce 2 x 2 x 2 x 2 = 16

#(b) Are the lines all parallel? If not, explain why not.

No all the lines are not parallel because because we have interaction terms which change the lines depending on the combinations of factors , this means there can be different lines for different situations.

#14. Now output the summary of your final model showing the estimated coefficients, and use that to write ˆ fi with all the estimated coefficients replacing the ˆ j pronumerals.

```
summary(bsfit_2$fit)
```

```
##
## Call:
## stats::glm(formula = recommend ~ age + malfunction + multi_purch +
##     ses + age:malfunction + multi_purch:ses, family = stats::binomial,
##     data = data)
##
```

```
## Coefficients:
##                           Estimate Std. Error z value Pr(>|z|)
## (Intercept)               0.043749   0.052270   0.837 0.402602
## age                      -0.050557   0.001241 -40.726  < 2e-16 ***
## malfunctionTRUE          -2.594165   0.830023  -3.125 0.001776 **
## multi_purchTRUE           3.002404   0.052501  57.188  < 2e-16 ***
## seslow                    0.164957   0.044534   3.704 0.000212 ***
## sesmid                   -0.034466   0.046306  -0.744 0.456694
## age:malfunctionTRUE      -0.118069   0.033164  -3.560 0.000371 ***
## multi_purchTRUE:seslow    0.486997   0.076185   6.392 1.63e-10 ***
## multi_purchTRUE:sesmid    0.070594   0.073503   0.960 0.336843
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 48075  on 39999  degrees of freedom
## Residual deviance: 29600  on 39991  degrees of freedom
## AIC: 29618
##
## Number of Fisher Scoring iterations: 10
```

$\hat{f}_i = 0.0437 - 0.0506 \times age - 2.5941 \times malfunction + 3.0024 \times multi_purch + 0.1645 \times seslow - 0.03445 \times sesmid - 0.1180 \times (age \times m$

#15. What is our estimate for the log-odds for a respondent: (a) who has a low Socio-Economic Status, yet purchased several Gadgets and none of them stopped working?

$$log_odds = 0.0437 - 0.0506 + 3.00 \times x_1 + 0.1650 \times x_2 + 0.4870 \times x_1 \times x_2)$$

$$log_odds = 0.0437 - 0.0506 + 3 \times 1 + 0.1650 \times 1 + 0.4870 \times 1 = 3.64$$

#Q15b who has a mid-range Socio-Economic Status, only purchased a single Gadget and it broke ses_mid +

$$log_odds = 0.0692 - 0.0506 - 2.4685$$

#Q16 Now apply your final model to the testing data. Produce a new tibble containing the predicted class and the prediction probabilities. Output the first 10 lines of this tibble.

```
bsr_2 <- logistic_reg() %>% set_engine("glm")
bsfit_2 <- bsr_2 %>%
  fit(recommend ~age+malfunction+multi_purch+ses+age:malfunction+multi_purch:ses, data = data_test)

#predicting values
predicted_class <- predict(bsfit_2, new_data = data_test)
predicted_class
```

```
## # A tibble: 10,000 x 1
##    .pred_class
##    <fct>
##  1 no
```

```
##  2 no
##  3 no
##  4 no
##  5 no
##  6 no
##  7 no
##  8 yes
##  9 no
## 10 yes
## # i 9,990 more rows
```

```
#predicting probability values
predicted_probability <- predict(bsfit_2, new_data = data_test, type = "prob")
predicted_probability
```

```
## # A tibble: 10,000 x 2
##    .pred_no .pred_yes
##       <dbl>     <dbl>
##  1    0.828  0.172
##  2    0.963  0.0374
##  3    1.00   0.000179
##  4    0.971  0.0293
##  5    1.00   0.000370
##  6    0.882  0.118
##  7    0.867  0.133
##  8    0.169  0.831
##  9    0.750  0.250
## 10    0.244  0.756
## # i 9,990 more rows
```

```
pr_tibble <-  cbind(predicted_class,predicted_probability)

head(pr_tibble,10)
```

```
##    .pred_class  .pred_no    .pred_yes
## 1           no 0.8281466 0.1718533770
## 2           no 0.9626068 0.0373931865
## 3           no 0.9998209 0.0001790631
## 4           no 0.9707457 0.0292543411
## 5           no 0.9996304 0.0003695867
## 6           no 0.8816751 0.1183249407
## 7           no 0.8672897 0.1327102938
## 8          yes 0.1694628 0.8305372473
## 9           no 0.7497134 0.2502865657
## 10         yes 0.2437082 0.7562917928
```

#Q17Now we need to evaluate our model. (a) Find the confusion matrix.(b) If leaving a review is classified as a success, find the sensitivity and specificity of our model.(c) Plot the ROC curve.(d) What is the AUC of this ROC curve?

#a

```
new_tibble <- cbind(true=data_test$recommend, pr_tibble)
new_tibble%>% conf_mat(truth=true, estimate=.pred_class)
```

```
##           Truth
## Prediction   no  yes
##        no  6640  978
##        yes  514 1868
```

#17b #sensitivity

```
my_sens <- 1868/(978+1868)
my_sens
```

```
## [1] 0.6563598
```

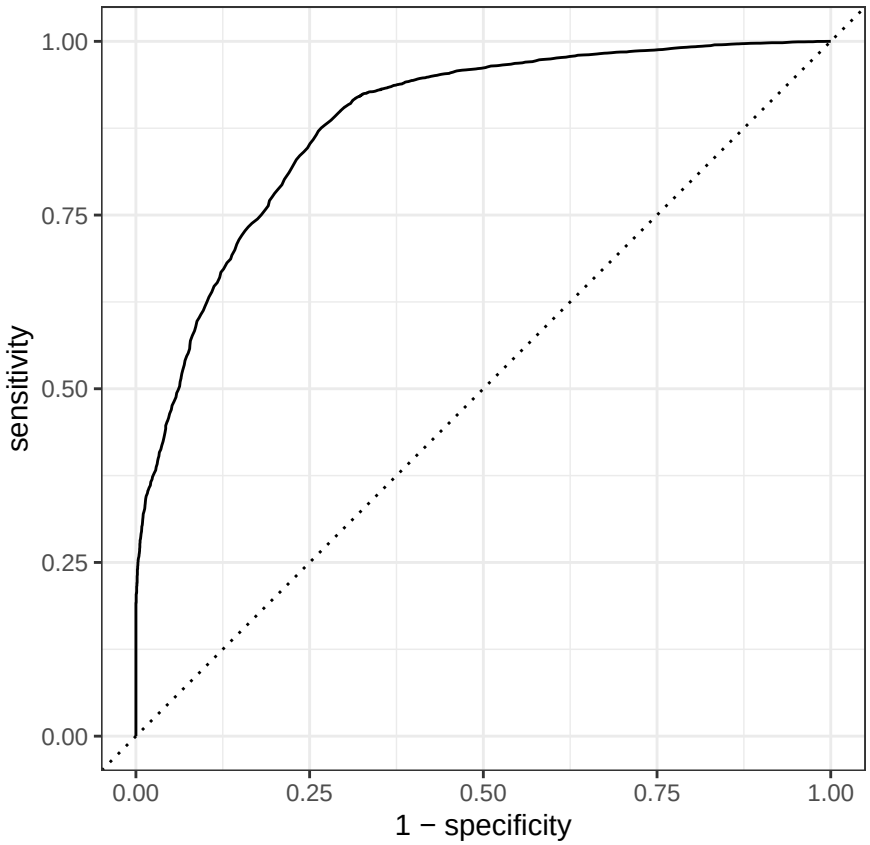
#specificity

```
my_spec <-  6640/(514+6640)
my_spec
```

```
## [1] 0.9281521
```

#Q17c

```
new_tibble %>% roc_curve(.pred_no, truth = true) %>% autoplot()
```



#17d

```
auc = new_tibble %>% roc_auc(.pred_no, truth = true)
auc
```

```
## # A tibble: 1 x 3
##   .metric .estimator .estimate
##   <chr>   <chr>          <dbl>
## 1 roc_auc binary         0.883
```

#Q18 #They have asked you to use the data to predict whether the Mayor will recommend the Gadget 2®. - 0.883 is the values given by AUC, by which we can say that the model is predicting good which means 88% of respondants will recommend it or not, for which other factors might also play some role.

The Mayor is 45 years old, and is known to have purchased many of the original Gadgets®, and still has them all in good working condition. He is known to be a prolific social media user, and has been spotted many times wearing a jacket with the company's logo on it

```
mr_mayor = tibble( age=45,malfunction="FALSE",multi_purch="TRUE",ses="high",company_aware="TRUE",social_

mayor_predict = predict(bsfit$fit, new_data=mr_mayor)
```